1. Use Gaussian elimination and pivoting technique to solve

$$1.19x_1 + 2.11x_2 - 100x_3 + x_4 = 1.12$$

$$14.2x_1 - 0.112x_2 + 12.2x_3 - x_4 = 3.44$$

$$100x_2 - 99.9x_3 + x_4 = 2.15$$

$$15.3x_1 + 0.110x_2 - 13.1x_3 - x_4 = 4.16$$

```
增廣矩陣:
[[   1.19     2.11  -100.       1.       1.12 ]
 [  14.2     -0.112   12.2     -1.       3.44 ]
 [   0.     100.     -99.9      1.       2.15 ]
 [  15.3      0.11   -13.1     -1.       4.16 ]]

上三角:
[[ 15.3          0.11        -13.1          -1.           4.16      ]
 [  0.         100.          -99.9           1.           2.15      ]
 [  0.           0.         -96.88176811   1.05676333   0.75126339]
 [  0.           0.           0.           0.1936057   -0.22908671]]

 x:
[ 0.17677633  0.0126921  -0.0206612  -1.18326429]

 A*x:
[1.12 3.44 2.15 4.16]
 b:
[1.12 3.44 2.15 4.16]
```

2. Find the inverse of the matrix A where

$$A = \begin{bmatrix} 4 & 1 & -1 & 0 \\ 1 & 3 & -1 & 0 \\ -1 & -1 & 6 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix}$$

```
 A:
[[ 4.  1. -1.  0.]
 [ 1.  3. -1.  0.]
 [-1. -1.  6.  2.]
 [ 0.  0.  2.  5.]]

增廣矩陣 [A|I]:
[[ 4.  1. -1.  0.  1.  0.  0.  0.]
 [ 1.  3. -1.  0.  0.  1.  0.  0.]
 [-1. -1.  6.  2.  0.  0.  1.  0.]
 [ 0.  0.  2.  5.  0.  0.  0.  1.]]

gaussian - jordan後的矩陣:
[[ 1.          0.          0.          0.          0.27969349 -0.08045977
   0.03831418 -0.01532567]
 [ 0.          1.          0.          0.         -0.08045977  0.37931034
   0.05747126 -0.02298851]
 [ 0.          0.          1.          0.          0.03831418  0.05747126
   0.21072797 -0.08429119]
 [ 0.          0.          0.          1.         -0.01532567 -0.02298851
  -0.08429119  0.23371648]]

inverse matrix A^(-1):
[[ 0.27969349 -0.08045977  0.03831418 -0.01532567]
 [-0.08045977  0.37931034  0.05747126 -0.02298851]
 [ 0.03831418  0.05747126  0.21072797 -0.08429119]
 [-0.01532567 -0.02298851 -0.08429119  0.23371648]]
```

3. Use Crout factorization for a tri-diagonal system to solve the problem

$$\begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{Bmatrix}.$$

```
 A:
[[ 3. -1.  0.  0.]
 [-1.  3. -1.  0.]
 [ 0. -1.  3. -1.]
 [ 0.  0. -1.  3.]]

 b:
[2. 3. 4. 1.]

L matrix:
[[ 3.          0.          0.          0.        ]
 [-1.          2.66666667  0.          0.        ]
 [ 0.         -1.          2.625       0.        ]
 [ 0.          0.         -1.          2.61904762]]

U matrix:
[[ 1.         -0.33333333  0.          0.        ]
 [ 0.          1.         -0.375       0.        ]
 [ 0.          0.          1.         -0.38095238]
 [ 0.          0.          0.          1.        ]]

 x:
[1.43636364 2.30909091 2.49090909 1.16363636]

 A*x:
[2. 3. 4. 1.]
 b:
[2. 3. 4. 1.]
```